

Errata: *Generalizing geometric nonwindowed scattering transforms on compact Riemannian manifolds*

Everything below is keyed to the published version, *Sampling Theory, Signal Processing, and Data Analysis* **22**, 19 (2024), <https://doi.org/10.1007/s43670-024-00093-x>, and numbering is that of the published article. The earlier preprint arXiv:2402.10771 is an initial draft and numbers results differently.

Items **A1–A5** change what is stated; **B** lists proof corrections that leave statements intact; **C** lists typos and is set in smaller type. None of the corrections changes the range $1 < q \leq 2$ or the conclusions of the paper.

This errata was prepared by Claude Opus 5.0. The authors have checked the corrections and take responsibility for all results.

A. Statement changes

A1. Theorem 1 holds modulo the constant component

Theorem 1 states $\sum_{j \in \mathbb{Z}} \|f * \psi_j\|_2^2 = C^2 \|f\|_2^2$. The telescoping in the proof gives $\sum_j |\hat{\psi}_j(k)|^2 = C^2$ only when $\lambda_k > 0$. For $k = 0$ we have $\lambda_0 = 0$, so

$$\hat{\psi}_j(0) = [|G(0)|^2 - |G(0)|^2]^{1/2} = 0 \quad \text{for every } j \in \mathbb{Z},$$

and the corresponding sum is 0 rather than C^2 . Hence the correct statement is

$$\sum_{j \in \mathbb{Z}} \|f * \psi_j\|_2^2 = C^2 \left(\|f\|_2^2 - |\hat{f}(0)|^2 \right) \leq C^2 \|f\|_2^2,$$

with equality precisely when $\hat{f}(0) = 0$. Only the inequality is used later in the paper, so no subsequent result is affected. The same correction applies to the display asserting $\|Wf\|_{\mathbf{L}^2_{\ell^2(\mathbb{Z})}(\mathcal{M})} = C\|f\|_2$ in Section 3, which restates Theorem 1.

The remark following Theorem 1 should be read accordingly: for $G(\lambda) = e^{-\lambda}$ one has $C = 1$ and the relation is an isometry on the functions with $\hat{f}(0) = 0$, not on all of $\mathbf{L}^2(\mathcal{M})$.

A2. Corollary 10 is an inequality

Corollary 10 states $\|\bar{S}_2^m f\|_{\ell^2(\mathbb{Z}^m)}^2 = C^{2m} \|f\|_{\mathbf{L}^2(\mathcal{M})}^2$. This should read $\leq$. Beyond the defect in A1 at the first layer, equality fails strictly at every later layer: the input to layer $k+1$ is $U[j_1, \dots, j_k]f \geq 0$, which has strictly positive integral whenever it does not vanish. Applying A1 one layer at a time gives, for instance,

$$\sum_{j_2 \in \mathbb{Z}} \|U[j_2]U[j_1]f\|_2^2 = C^2 \left(\|U[j_1]f\|_2^2 - \text{vol}(\mathcal{M})^{-1} \|U[j_1]f\|_1^2 \right),$$

where the subtracted term is the square of the corresponding $q = 1$ scattering coefficient.

A3. The regularity condition should be $c_2/c_1 < 2$

Lemma 4, and the statements that quote it (Theorem 6, Corollary 8, Theorem 12, Corollary 13, Theorem 14, Theorem 18), assume $c_1 c_2 < 2$. The condition that the proof requires is

$$\frac{c_2}{c_1} < 2.$$

In the proof of Lemma 4, item 2 of Lemma 3 gives $|y - z| \leq c_1^{-1} r(y, z)$, so

$$r(y, w_x) \leq c_2 |y - w_x| \leq c_2 |y - z| \leq \frac{c_2}{c_1} r(y, z),$$

and with $r(y, z) \leq \frac{1}{2} r(x, y)$ the conclusion needs $1 - \frac{c_2}{2c_1} > 0$. The remaining occurrences of $c_1 c_2$ in that proof should be replaced by c_2/c_1 in the same way.

The distinction matters beyond bookkeeping. The product $c_1 c_2$ is not invariant under rescaling the chart maps φ_i : replacing φ_i by $a\varphi_i$ sends $c_1 c_2$ to $c_1 c_2/a^2$, so the stated condition could be met by reparametrizing. The ratio c_2/c_1 is invariant, satisfies $c_2/c_1 \geq 1$ always, and measures the distortion of the metric in the coordinates of the covering.

Example 1 is unaffected: there $c_1 = 1$, so $c_2/c_1 = c_2$ and both computations still give a value below 2.

A4. The Schwartz condition belongs to the wavelet generator

Theorem 6, Corollary 8, Theorem 12, Corollary 13 and Theorem 14 assume $G \in \mathcal{S}(\mathbb{R}^+)$, but the hypothesis that Lemma 2 actually requires is on the function generating the wavelets. Writing

$$F(u) := [|G(u/2)|^2 - |G(u)|^2]^{1/2}, \quad \hat{\psi}_j(n) = F(2^j \lambda_n),$$

Lemma 2 applies when $F \in \mathcal{S}(\mathbb{R}^+)$, and $F(0) = 0$ holds automatically. The condition $G \in \mathcal{S}(\mathbb{R}^+)$ does not imply it: the difference may vanish to first order at the origin, in which case F behaves like $u^{1/2}$ and is not differentiable at $u = 0$. This happens for $G(u) = e^{-u}$, where $F(u)^2 = e^{-u} - e^{-2u} = u + O(u^2)$, which is exactly the choice named in the remark after Theorem 1.

A sufficient condition is that $|G(u/2)|^2 - |G(u)|^2 = u^2 h(u)$ for some $h \in \mathcal{S}(\mathbb{R}^+)$ that is positive on $[0, \infty)$, since then $F(u) = u h(u)^{1/2}$. The Gaussian $G(u) = e^{-u^2}$ satisfies this: $e^{-u^2/2} - e^{-2u^2}$ is even, vanishes to second order at the origin and is positive on $(0, \infty)$.

A5. Missing hypothesis and constants in the stability theorems

Theorem 16 is stated for $f \in \mathbf{L}^2(\mathcal{M})$, but the bound contains λ^n and the proof invokes Lemma 15; the hypothesis that f be λ -bandlimited is needed and should be added to the statement.

In both Theorem 16 and Theorem 18 the constant depends on the number of layers. Theorem 9 contributes a factor C^m and Theorem 12 a factor $C_q^{m/q}$, so the constants should be written $C_m(\mathcal{M})$ and $C_{m,q}(\mathcal{M})$. In the proof of Theorem 18, the displayed constant C_q should be $C_q^{m/q}$.

B. Proof corrections (statements unchanged)

B1. The inductions in Theorems 9 and 12

Both proofs reach a sum of *norms of differences* and then identify it with a scattering norm, which is a sum of *differences of norms*. In Theorem 9 the step reads

$$C^2 \sum_{(j_1, \dots, j_k) \in \mathbb{Z}^k} \|U[j_1, \dots, j_k]f - U[j_1, \dots, j_k]g\|_2^2 = C^2 \|\bar{S}_2^k f - \bar{S}_2^k g\|_{\ell^2(\mathbb{Z}^k)}^2,$$

and Theorem 12 has the analogous line with $\|\cdot\|_q$. These two quantities are not equal, and the reverse triangle inequality bounds the second by the first, so the induction hypothesis does not cover the quantity produced by the step.

Correction. Run the induction on the larger quantity. For $m \geq 1$ set

$$E_m(f, g) := \sum_{p \in \mathbb{Z}^m} \|U[p]f - U[p]g\|_2^2, \quad E_m^{(q)}(f, g) := \left(\sum_{p \in \mathbb{Z}^m} \|U[p]f - U[p]g\|_q^2 \right)^{q/2},$$

so that the reverse triangle inequality gives $\|\bar{S}_2^m f - \bar{S}_2^m g\|_{\ell^2(\mathbb{Z}^m)}^2 \leq E_m(f, g)$ and $\|\bar{S}_q^m f - \bar{S}_q^m g\|_{\ell^2(\mathbb{Z}^m)}^q \leq E_m^{(q)}(f, g)$. Since the modulus is nonexpansive,

$$\begin{aligned} E_{k+1}(f, g) &= \sum_{p \in \mathbb{Z}^k} \sum_{j_{k+1} \in \mathbb{Z}} \left| \|U[p]f * \psi_{j_{k+1}}\|_2 - \|U[p]g * \psi_{j_{k+1}}\|_2 \right| \\ &\leq \sum_{p \in \mathbb{Z}^k} \sum_{j_{k+1} \in \mathbb{Z}} \|(U[p]f - U[p]g) * \psi_{j_{k+1}}\|_2^2 \\ &\leq C^2 E_k(f, g), \end{aligned}$$

by Theorem 1 applied to $U[p]f - U[p]g$, with $E_0(f, g) = \|f - g\|_2^2$. The same computation with Corollary 8 in place of Theorem 1 gives $E_{k+1}^{(q)} \leq C_q E_k^{(q)}$. The stated bounds follow. The chains already written in the two proofs are otherwise unchanged.

B2. Smaller items in the proof of Theorem 6

- In the display bounding the case $2r_i \geq \delta$, the middle expression carries $(1 + 2^j r(x, x_i))^{n+1}$; the exponent should be $2^{j/2}$, as in the lines before and after it.
- In the conclusion of the estimate, $\|Wb_i\|_{\mathbf{L}_{\ell^2(\mathbb{Z})}^1(B(x_i, r_i)^c)}$ should be $\|Wb_i\|_{\mathbf{L}_{\ell^2(\mathbb{Z})}^1(B(x_i, 2r_i)^c)}$, matching the exceptional set used throughout.

B3. Lemma 17

The displayed chain bounds $|f(x) - V_\xi f(x)|$ pointwise, while the statement is an $\mathbf{L}^q(\mathcal{M})$ bound; the passage from one to the other contributes a factor $\mu(\mathcal{M})^{1/q}$, which is absorbed into $C_{q, \text{Vol}(\mathcal{M})}$ but is not shown.

C. Typos

Section 2

- Notation: “the maximum placement of γ ” should be displacement; “We use the notation $\text{Isom}(\mathcal{M}_1, \mathcal{M}_2)$ be the set” is missing a verb.
- §2.2: “a notation of translation does not necessarily exist, To motivate...” should be “a notion of translation does not necessarily exist. To motivate...”. In the same passage, the basis $\{c_n(x)\}_{n \geq 0}$ should be $\{e_n(x)\}_{n \geq 0}$, “fourier” should be capitalized, and the display stops short of the point being made, namely $\langle f * g, e_n \rangle = \hat{f}(n)\hat{g}(n)$.
- “since the the geometry of $\mathcal{M}$ ” has a repeated word.
- Equation (5): $|G(2^{j-1}n)|^2 - |G(2^j n)|^2$ should read $|G(2^{j-1}\lambda_n)|^2 - |G(2^j \lambda_n)|^2$.
- Equation (12): the m -layer windowed propagator is written with the same symbol as the unwinded one; it should be $U_J[j_1, \dots, j_m]f := U_J[j_m] \cdots U_J[j_1]f$.
- The display before (13) is missing f on the left: $S_J[j_1, \dots, j_m]f = A_J U_J[j_1, \dots, j_m]f$. In (13) itself, S_j should be S_J .
- Equation (14): $S[j_1, \dots, j_m]f$ should be $S_J[j_1, \dots, j_m]f$, and the right-hand side carries a factor $G(0)$, which is 1 only under the normalization $C = 1$.
- Contributions list: “arbitrary”.

Sections 3–5

- §3, definition of W : the kernel is given as $\vec{K} = \{K_{2^{-j/2}}\}_{j \in \mathbb{Z}}$. Since $\hat{\psi}_j(n) = F(2^j \lambda_n)$ with F as in A4, and K_t carries the multiplier $F(t^2 \lambda_k)$, the scale matching ψ_j is $t = 2^{j/2}$; as printed, the j -th entry of Wf is $f * \psi_{-j}$. The estimates in the proof of Theorem 6 are written consistently with $t = 2^{-j/2}$, so correcting the kernel index also sends $j \mapsto -j$ there: each $2^{nj/2}$ becomes $2^{-nj/2}$, each $2^{j/2}$ becomes $2^{-j/2}$, and the split becomes $j \leq 0$ and $j > 0$. Nothing changes numerically, since every occurrence of W is summed over $j \in \mathbb{Z}$.
 - Lemma 2: the letter n denotes both the dimension of $\mathcal{M}$ and the summation index of $K_t(x, y) = \sum_n F(t^2 \lambda_n) e_n(x) \overline{e_n(y)}$, while the bound uses t^{-n} and the exponent $n + 1$ in the first sense.
 - §5: the λ -bandlimited condition is written $\hat{f}(k) = \langle f, \phi_k \rangle = 0$; the eigenfunctions are e_k , and ϕ_j already denotes the low-pass filter with $\hat{\phi}_j(n) = G(2^j \lambda_n)$.
 - Section 3 title and Theorem 11: “littlewood paley” should be “Littlewood–Paley”.
 - “Caulderon-Zygmund” in the proof of Theorem 6, and “Holder” in the proof of Lemma 17.
 - After (13): “stable to diffeomorphisms”.
 - Theorem 9, proof: the delimiters in $||f * \psi_j||_2 - |g * \psi_j||_2|^2$ are mismatched; ψ_{j+1} should be $\psi_{j_{k+1}}$ in two places; $U[j_1, \dots, j_n]g$ should be $U[j_1, \dots, j_k]g$; and the hypothesis is assumed for n but then invoked “for some $k \geq 1$ ”.
 - Theorem 12, proof: $\ell^2(\mathbb{Z}^m)$ should be $\ell^2(\mathbb{Z})$ in the $m = 1$ display and $\ell^2(\mathbb{Z}^{m+1})$ in the $m + 1$ display; the line beginning “= C_q ” should be an inequality; and in the final display $\|\cdot\|_{\mathbf{L}_{\ell^2(\mathbb{Z})}^q(\mathcal{M})}$ should be $\|\cdot\|_{\mathbf{L}^q(\mathcal{M})}$.
 - Conclusion: “manifolds that are not restricted the conditions” is missing “to”.
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D. Unaffected

The framework of Sections 2 and 3, the weak $(1, 1)$ estimate of Theorem 6 and its interpolation in Corollary 8, the isometry invariance of Theorems 11 and 14, and the boundedness and stability results for $q \in (1, 2]$ all stand as stated, with the hypotheses as corrected in A3 and A4.